Rational Expressions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(5)$.

Final Answer: 21. Substitute $x = 5$: $f(5) = 3(5) + 6 = 21$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 7$ in operations with radical expressions, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 4(2) + 7 = 15$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for operations with radical expressions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 5x + 8$ in graphs and characteristics of square root functions, evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 5(2) + 8 = 18$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for graphs and characteristics of square root functions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 6x + 9$ in graphs and characteristics of cube root functions, evaluate $f(2)$.

Final Answer: 21. Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for graphs and characteristics of cube root functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Simplify the complex rational expression: $\frac{\frac{1}{x}-\frac{1}{y}}{\frac{1}{x^2}-\frac{1}{y^2}}$, where $xeqy$ and $xyeq0$.

Factor the denominator as a difference of squares: $\frac{1}{x^2}-\frac{1}{y^2}=\left(\frac{1}{x}-\frac{1}{y}\right)\left(\frac{1}{x}+\frac{1}{y}\right)$. Cancel the common factor $\left(\frac{1}{x}-\frac{1}{y}\right)$: $\frac{1}{\frac{1}{x}+\frac{1}{y}}=\frac{1}{\frac{y+x}{xy}}=\frac{xy}{x+y}$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · FoundationsOn Section I Part B of the AP Precalculus exam, a student allocates 50 minutes to answer 12 questions using a graphing calculator. What is the average time per question available in minutes, rounded to the nearest tenth?

Calculate the ratio of total time to questions: $\frac{50\text{ min}}{12\text{ questions}}=\frac{25}{6}\approx 4.167\text{ min/question}$. Rounded to the nearest tenth, this is 4.2 minutes per question.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the rational function $r(x)=\frac{2x^2-2x-12}{x^2-9}$, find: (a) the domain, (b) the coordinates of any removable discontinuity (hole), (c) the vertical and horizontal asymptotes, and (d) the x- and y-intercepts.

Step 1: Factor numerator and denominator completely: $r(x)=\frac{2(x^2-x-6)}{(x-3)(x+3)}=\frac{2(x-3)(x+2)}{(x-3)(x+3)}$. Step 2: Answer each part:(a) Domain: $(-\infty,-3)\cup(-3,3)\cup(3,\infty)$, i.e. $xeq\pm3$.(b) Removable Discontinuity: The factor $(x-3)$ cancels; for $xeq3$, $r(x)=\frac{2(x+2)}{x+3}$. Evaluating at $x=3$ gives $y=\frac{2(5)}{6}=\frac{5}{3}$. Hole at $\left(\mathbf{3,}\ \mathbf{\frac{5}{3}}\right)$.(c) Asymptotes: Vertical asymptote at $\mathbf{x=-3}$; ho.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2-4xy+8y^2=36$. Find the angle of rotation θ, standard form in the rotated system (x',y'), and the lengths of the semi-major and semi-minor axes.

Step 1: Determine the rotation angle θ: Here $A=5$, $B=-4$, and $C=8$.
$\cot(2\theta)=\frac{A-C}{B}=\frac{5-8}{-4}=\frac{-3}{-4}=\frac{3}{4}>0$. Since $\cot(2\theta)=3/4$, by right triangle trigonometry $\cos(2\theta)=\frac{3}{5}$.
Using half-angle formulas for $\theta\in(0,\pi/4)$: $\cos\theta=\sqrt{\frac{1+\cos(2\theta)}{2}}=\sqrt{\frac{1+3/5}{2}}=\sqrt{\frac{4}{5}}=\frac{2}{\sqrt{5}},$.

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12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2+1}{x^2-4}$. Describe its end behavior using limits.

Step 1: Domain and vertical asymptotes: The denominator factors as $x^2 - 4 = (x-2)(x+2)$. The numerator $3x^2 + 1 > 0$ has no real zeros. Thus, there are no common factors and no removable holes. The vertical asymptotes are $x = 2$ and $x = -2$. Step 2: Horizontal asymptote and end behavior: Numerator and denominator have equal degree $n = d = 2$: $\lim_{x\to\pm\infty} f(x) = \lim_{x\to\pm\infty} \frac{3+1/x^2}{1-4/x^2} = \frac{3}{1} = 3.$ The horizontal asymptote is $y = 3$. Notice tha.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Solve the rational inequality $\frac{x+4}{x-2} \le 3$ and express the solution set in interval notation.

Step 1: Subtract 3 and combine over a single common denominator: $\frac{x+4}{x-2} - 3 \le 0 \Rightarrow \frac{(x+4)-3(x-2)}{x-2} \le 0 \Rightarrow \frac{-2x+10}{x-2} \le 0.$ Step 2: Factor and identify critical boundary values: $\frac{-2(x-5)}{x-2} \le 0 \Leftrightarrow \frac{x-5}{x-2} \ge 0.$ Critical values are $x = 2$ (undefined) and $x = 5$ (zero). Step 3: Sign chart analysis: The quotient $\frac{x-5}{x-2}$ is positive on $(-\infty, 2)$ and $(5, \infty)$, and negative on.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Determine the slant asymptote of $g(x) = \frac{2x^3-3x^2+5}{x^2+1}$ and find the coordinates of any point where the curve intersects this asymptote.

Since $\deg(\text{num}) = 3$ and $\deg(\text{denom}) = 2$ with $3 - 2 = 1$, a slant asymptote exists. Perform polynomial long division: $(2x^3 - 3x^2 + 5) \div (x^2 + 1) = (2x - 3) + \frac{-2x+8}{x^2+1}.$ The linear quotient gives the slant asymptote: $y = 2x - 3$. To find where the graph intersects this asymptote, set the remainder equal to zero: $\frac{-2x+8}{x^2+1} = 0 \Rightarrow -2x + 8 = 0 \Rightarrow x = 4.$ Substitute $x = 4$ into the asymptote equation: $y = 2(4) - 3 = 5.$ The curve intersects.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 9.01 Trigonometric Functions**

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2+5x-3}{x^2+4x+3}$.

Factor numerator and denominator completely: $R(x) = \frac{(2x-1)(x+3)}{(x+1)(x+3)}.$ The factor $(x + 3)$ cancels for $xeq -3$. Evaluate the simplified form $g(x) = \frac{2x-1}{x+1}$ at $x = -3$: $g(-3) = \frac{2(-3)-1}{-3+1} = \frac{-7}{-2} = 3.5.$ Key features:Hole: $(-3, 3.5)$Vertical asymptote: $x + 1 = 0 \Rightarrow x = -1$Horizontal asymptote: Equal degrees (2 and 2) $\Rightarrow y = 2/1 = 2x$-intercept: Numerator root $2x - 1 = 0 \Rightarrow (0.5, 0)y$-intercept.

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16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 1.01 Functions & Their Graphs**

Find all real solutions to the rational equation $\frac{x}{x-2}+\frac{1}{x+3}=\frac{5}{x^2+x-6}$.

Notice that the common denominator factors as $x^2+x-6=(x-2)(x+3)$. Domain restrictions: $xeq2$ and $xeq-3$. Multiply both sides by $(x-2)(x+3)$: $x(x+3)+1(x-2)=5\Rightarrow x^2+3x+x-2=5$ $x^2+4x-7=0$ Apply the quadratic formula with $a=1, b=4, c=-7$:

$x=\frac{-4\pm\sqrt{4^2-4(1)(-7)}}{2(1)}=\frac{-4\pm\sqrt{16+28}}{2}=\frac{-4\pm\sqrt{44}}{2}=\frac{-4\pm2\sqrt{11}}{2}=\mathbf{-2\pm\sqrt{11}}$ Neither solution equals 2 or -3, so both solutio.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 10.01 Matrices & Systems**

Decompose the rational expression $R(x)=\frac{7x-1}{x^2-x-6}$ into partial fractions using the Heaviside cover-up method.

Step 1: Factor denominator: $x^2-x-6=(x-3)(x+2)$. Set up the decomposition: $\frac{7x-1}{(x-3)(x+2)}=\frac{A}{x-3}+\frac{B}{x+2}$. Step 2: Apply Heaviside cover-up method:For A (evaluate at $x=3$): $A=\frac{7x-1}{x+2}\Big|_{x=3}=\frac{7(3)-1}{3+2}=\frac{20}{5}=4$. For B (evaluate at $x=-2$): $B=\frac{7x-1}{x-3}\Big|_{x=-2}=\frac{7(-2)-1}{-2-3}=\frac{-15}{-5}=3$. The partial fraction decomposition is $\frac{4}{x-3}+\frac{3}{x+2}$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 11.01 Probability & Statistics**

Find the equation of the slant (oblique) asymptote of the rational function $g(x)=\frac{3x^2-5x+7}{x-2}$, and compute $\lim_{x\to2^+}g(x)$.

Step 1: Perform polynomial long division (or synthetic division by $c=2$): $3x^2-5x+7=(x-2)(3x+1)+9\Rightarrow g(x)=3x+1+\frac{9}{x-2}$. Step 2: Identify the slant asymptote: As $x\to\pm\infty$, $\frac{9}{x-2}\to0$, so the slant asymptote is $\mathbf{y=3x+1}$. Step 3: One-sided limit at $x=2^+$: As $x\to2^+$, $x-2\to0^+$ and numerator $\to9>0$, so $\lim_{x\to2^+}g(x)=\mathbf{+\infty}$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 2.01 Quadratics & Complex Numbers**

Convert the repeating decimal $x=0.4\overline{36}=0.4363636\ldots$ into an irreducible rational fraction $\frac{p}{q}$.

Let $x=0.4\overline{36}$. Multiply by powers of 10 to shift the non-repeating and repeating parts: $10x=4.\overline{36}$ $1000x=436.\overline{36}$ Subtract the first equation from the second to eliminate the repeating tails: $1000x-10x=436.\overline{36}-4.\overline{36}\Rightarrow990x=432$ $x=\frac{432}{990}$ Divide numerator and denominator by their greatest common factor (18): $432\div18=24,\qquad990\div18=55\Rightarrow x=\mathbf{\frac{24}{55}}$

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Polynomial Functions

Level 3 · AdvancedAn invasive plant covers an area modeled by $A(t) = 120(1.15)^t$ square meters, where t is measured in weeks. Write the equation setup to determine when the area reaches 500 square meters, and find the value of t to three decimal places with correct units.

Setup: Write the equation before numerical solving: $120(1.15)^t = 500 \Rightarrow (1.15)^t = \frac{500}{120} = \frac{25}{6}$. Solve: Take the natural logarithm: $t\ln(1.15) = \ln\left(\frac{25}{6}\right) \Rightarrow t = \frac{\ln(25/6)}{\ln(1.15)}$. Evaluating on the calculator: $t \approx \frac{1.427116}{0.139762} \approx 10.211$ weeks. Final answer: 10.211 weeks.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 4$ in review: laws of exponents, evaluate $f(2)$.

a. 16 ✓ **b.** 22

c. 12 **d.** 20

Substitute $x = 2$: $f(2) = 6(2) + 4 = 16$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 5$ in fractional powers, evaluate $f(2)$.

a. 9 ✓ **b.** 11

c. 4 **d.** 13

Substitute $x = 2$: $f(2) = 2(2) + 5 = 9$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for review: laws of exponents consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5 **b.** 17.5

c. 18.5 ✓ **d.** 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for fractional powers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the rational equation $\frac{x+1}{x-3} + \frac{k}{x+1} = \frac{x+13}{x^2-2x-3}$ where k is a constant. What is the sum of all real values of k for which the equation has no real solution?

a. -10 ✓

b. -7

c. -3

d. 10

Step 1: Determine the restricted values from the denominators. The denominators are $x - 3$, $x + 1$, and $x^2 - 2x - 3 = (x - 3)(x + 1)$. Thus, the domain of the equation is all real numbers except $x = 3$ and $x = -1$. Step 2: Clear the denominators by multiplying both sides by $(x - 3)(x + 1)$: $(x + 1)^2 + k(x - 3) = x + 13$ Step 3: Expand and rewrite as a standard quadratic equation: $(x^2 + 2x + 1) + kx - 3k = x + 13$ $x^2 + (k + 1)x - (3k + 12) = 0$ Step 4: Factor the quadratic equation. Notice that substitu.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The graph of the rational function $g(x) = \frac{ax^2 + bx + 1}{x^2 - 4}$ has a horizontal asymptote at $y = 2$ and a vertical asymptote at $x = 2$. Given that $g(0) = -1/4$, find the value of a.

a. 1

b. **2** ✓

c. 3

d. 4

The horizontal asymptote of a rational function $g(x) = \frac{P(x)}{Q(x)}$ where the degree of $P(x)$ is equal to the degree of $Q(x)$ is the ratio of the leading coefficients. In this case, the leading coefficient of the numerator is a and the leading coefficient of the denominator is 1. Therefore, the horizontal asymptote is $y = \frac{a}{1} = a$. We are given that the horizontal asymptote is $y = 2$, so $a = 2$. The vertical asymptotes occur at the zeros of the denominator that are not zeros of the numerator.

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the rational function $f(x) = \frac{x^2 - 2x + c}{x^2 + 2x + c}$ where c is a real constant. If the domain of f is the set of all real numbers and the maximum value of $f(x)$ is 4, what is the value of c?

a. $\frac{25}{9}$ ✓

b. $\frac{9}{4}$

c. 3

d. $\frac{16}{5}$

We are given the rational function $f(x) = \frac{x^2 - 2x + c}{x^2 + 2x + c}$. **Step 1: Domain constraint** For the domain of f to be all real numbers, the denominator $x^2 + 2x + c$ must have no real roots. Therefore, its discriminant must be strictly negative: $\Delta_{\text{denom}} = 2^2 - 4(1)(c) = 4 - 4c1$ **Step 2: Range and maximum value via discriminant method** Let $y = f(x)$. Because $x^2 + 2x + c > 0$ for all $x \in \mathbb{R}$, we can multiply through by the denominator.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which of the following is equivalent to the expression $\frac{x^2 - 25}{2x - 10}$ for all $xeq5$?

a. $\frac{x + 5}{2}$ ✓

b. $\frac{x - 5}{2}$

c. $x + 5$

d. $\frac{x + 5}{2(x - 5)}$

To simplify the rational expression $\frac{x^2 - 25}{2x - 10}$, factor both the numerator and the denominator completely: 1. The numerator is a difference of squares: $x^2 - 25 = (x - 5)(x + 5)$ 2. The denominator has a common factor of 2: $2x - 10 = 2(x - 5)$ 3. Rewrite the expression with the factored forms: $\frac{(x - 5)(x + 5)}{2(x - 5)}$ 4. For $xeq5$, the common factor $(x - 5)$ in the numerator and denominator cancels out: $\frac{x + 5}{2}$ Therefore, the correct choice is $\frac{x + 5}{2}$. Common erro.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Which of the following expressions is equivalent to $\frac{x^2-9}{x-3}$ for all values of x where the expression is defined?

a. $x+3$ ✓

b. $x-3$

c. x^2-3

d. $x+9$

The expression $\frac{x^2-9}{x-3}$ can be simplified by factoring the numerator. The numerator x^2-9 is a difference of squares, which factors as $(x-3)(x+3)$. Therefore, the expression becomes $\frac{(x-3)(x+3)}{x-3}$. For values of x where $xeq3$, we can cancel the common factor $(x-3)$ from the numerator and the denominator, leaving $x+3$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the sum $\frac{1}{x(x+1)}+\frac{1}{(x+1)(x+2)}+\frac{1}{(x+2)(x+3)}$ using partial fractions.

a. $\frac{1}{x+3}$

b. $\frac{3}{(x+1)(x+2)}$

c. $\frac{3}{x(x+3)}$ ✓

d. $\frac{1}{x}-\frac{1}{x+3}$

Telescoping: $\left(\frac{1}{x}-\frac{1}{x+1}\right)+\left(\frac{1}{x+1}-\frac{1}{x+2}\right)+\left(\frac{1}{x+2}-\frac{1}{x+3}\right)=\frac{1}{x}-\frac{1}{x+3}=\frac{3}{x(x+3)}$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Write $\frac{x}{(x+1)^2}$ in partial fractions.

a. $\frac{1}{x+1}-\frac{1}{(x+1)^2}$ ✓

b. $\frac{x}{(x+1)^2}$ cannot be decomposed

c. $\frac{1}{x+1}+\frac{1}{(x+1)^2}$

d. $\frac{1}{(x+1)^2}$

Applying Rational Expressions & Functions principles, $x=A(x+1)+B \Rightarrow A=1, B=-1$. Therefore, the verified correct option is $\frac{1}{x+1}-\frac{1}{(x+1)^2}$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Determine if $f(x) = \frac{3x^2 - 5}{x^2 + 2}$ has any vertical asymptotes.

a. Yes at $x = 0$

b. Yes at $x = \pm\sqrt{2}$

c. Yes at $y = 3$

d. **No (denominator $x^2 + 2 > 0$ always)** ✓

Applying Rational Expressions & Functions principles, $x^2 + 2 \geq 2 > 0$: no real zeros. Therefore, the verified correct option is No (denominator $x^2 + 2 > 0$ always).

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What happens to $f(x) = \frac{1}{x}$ as $x \to 0^+$?

a. $f \to 0$

b. $f \to -\infty$

c. $f \to +\infty$ ✓

d. $f \to 1$

Applying Rational Expressions & Functions principles, From the right, $1/x \to +\infty$. Therefore, the verified correct option is $f \to +\infty$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which of these is a rational function?

a. $f(x) = \sqrt{x}$

b. $f(x) = \frac{x^2 - 1}{x + 3}$ ✓

c. $f(x) = x^{2/3}$

d. $f(x) = 2^x$

Applying Rational Expressions & Functions principles, Ratio of polynomials only. Therefore, the verified correct option is $f(x) = \frac{x^2 - 1}{x + 3}$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find all solutions of $\frac{x^2 + 2x}{x^2 - 1} = 1 + \frac{3}{x - 1}$.

a. $x = 1$

b. No solution

c. $x = -4$

d. $x = 4$ ✓

LCD $= (x - 1)(x + 1)$: $x^2 + 2x = (x^2 - 1) + 3(x + 1) \Rightarrow x^2 + 2x = x^2 + 3x + 2 \Rightarrow -x = 2 \Rightarrow x = -2$. Check: denom nonzero at $x = -2$. Answer: $x = -2$.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the partial fraction decomposition of $\frac{1}{x^3 - x}$.

a. $\frac{1}{x(x-1)} - \frac{1}{x+1}$

b. $-\frac{1}{x} + \frac{1/2}{x-1} + \frac{1/2}{x+1}$ ✓

c. $\frac{1}{x} + \frac{1}{x-1} + \frac{1}{x+1}$

d. $\frac{1}{x^2-1} - \frac{1}{x}$

$x^3 - x = x(x-1)(x+1)$. $1 = A(x^2-1) + Bx(x+1) + Cx(x-1)$. $x=0$: $-A = 1 \Rightarrow A = -1$. $x=1$: $2B = 1 \Rightarrow B = 1/2$. $x=-1$: $2C = 1 \Rightarrow C = 1/2$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the exact solution of $\frac{x}{x-1} + \frac{2}{x+1} = \frac{6}{x^2-1}$.

a. $x = -2$

b. No solution

c. $x = 3$

d. $x = 2$ ✓

Multiply by $(x-1)(x+1)$: $x(x+1) + 2(x-1) = 6 \Rightarrow x^2 + x + 2x - 2 = 6 \Rightarrow x^2 + 3x - 8 = 0 \Rightarrow x = \frac{-3 \pm \sqrt{41}}{2}$. Not clean.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 11.01 Probability & Statistics

Is $f(x) = \frac{1}{x^2 + 4}$ always positive?

a. No

b. Only for $x > 0$

c. Only for $x < 0$

d. **Yes** ✓

Applying Rational Expressions & Functions principles, $x^2 + 4 > 0$ always $\rightarrow f(x) > 0$ always. Therefore, the verified correct option is Yes.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the vertical asymptotes of $f(x) = \frac{x+1}{x^2-4}$.

a. $x = \pm 2$ ✓

b. No VA

c. $x = -1, \pm 2$

d. $x = 2$ only

Applying Rational Expressions & Functions principles, $x^2 - 4 = 0 \Rightarrow x = \pm 2$. Therefore, the verified correct option is $x = \pm 2$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.01 Polynomial Functions

Find the x-intercepts and vertical asymptotes of $f(x) = \frac{x^2-x-6}{x^2-4}$.

a. x-int: $x = 3, -2$ (but $x = -2$ is a hole); VA: $x = 2$

b. x-int: $x = 3$; VA: $x = \pm 2$

c. x-int: $x = 3, -2$; VA: $x = \pm 2$

d. **x-int:** $x = 3$**; VA:** $x = 2$ **only** ✓

Num: $(x-3)(x+2)$. Den: $(x-2)(x+2)$. Cancel $(x+2)$: hole at $x = -2$. VA at $x = 2$. x-int: set num of simplified function $= 0$: $x = 3$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Find the x-intercept of $f(x) = \frac{x-3}{x+2}$.

a. $x = -2$

b. $x = 2$

c. $x = 3$ ✓

d. $x = -3$

Applying Rational Expressions & Functions principles, Set numerator to zero: $x = 3$. Therefore, the verified correct option is $x = 3$.

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Radicals & Rational Exponents

The domain of $f(x) = \frac{1}{x-3}$ excludes:

a. $x = 3$ ✓

b. $x = 0$

c. All reals

d. $x = -3$

Applying Rational Expressions & Functions principles, Denominator cannot be zero. Therefore, the verified correct option is $x = 3$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Sequences & Series

Find the vertical asymptote of $f(x) = \frac{2}{x+5}$.

a. $x = -5$ ✓

b. $y = -5$

c. $x = 5$

d. $x = 0$

Applying Rational Expressions & Functions principles, Set denominator to zero. Therefore, the verified correct option is $x = -5$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Conic Sections

Find the inverse of $f(x) = \frac{3x+2}{x-1}$ and state its domain.

a. $f^{-1}(x) = \frac{x+2}{x-3}$; **domain: all reals except** $x = 3$ ✓

b. $f^{-1}(x) = \frac{3x-2}{x+1}$

c. $f^{-1}(x) = \frac{x+2}{x+3}$

d. $f^{-1}(x) = \frac{x-2}{x+3}$

Swap: $x = \frac{3y+2}{y-1} \Rightarrow x(y-1) = 3y+2 \Rightarrow xy - x = 3y+2 \Rightarrow y(x-3) = x+2 \Rightarrow y = \frac{x+2}{x-3}$.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Trigonometric Functions

Identify: $\frac{p(x)}{q(x)}$ is called a _____ function.

a. Irrational

b. **Rational ✓**

c. Polynomial

d. Transcendental

Applying Rational Expressions & Functions principles, By definition. Therefore, the verified correct option is Rational.

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